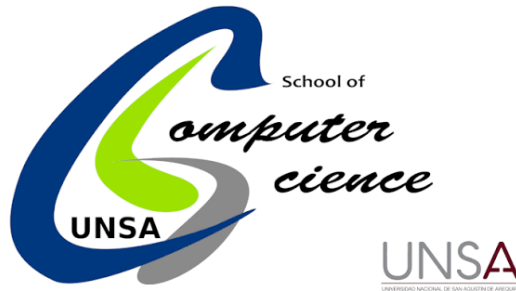


**“AÑO DE LA RECUPERACIÓN Y CONSOLIDACIÓN DE
LA ECONOMÍA PERUANA”.**



**ESCUELA PROFESIONAL DE CIENCIA DE LA
COMPUTACIÓN
CLOUD COMPUTING**

Implementación y demostración de la arquitectura de Kubernetes con una aplicación multicapa

Alumnos:

Carazas Quispe, Alessander Jesus Humpiri
Valdivia, Freddy Leonel Mariños Hilario,
Princece Yorwin Rivas Huanca, Diego Raul

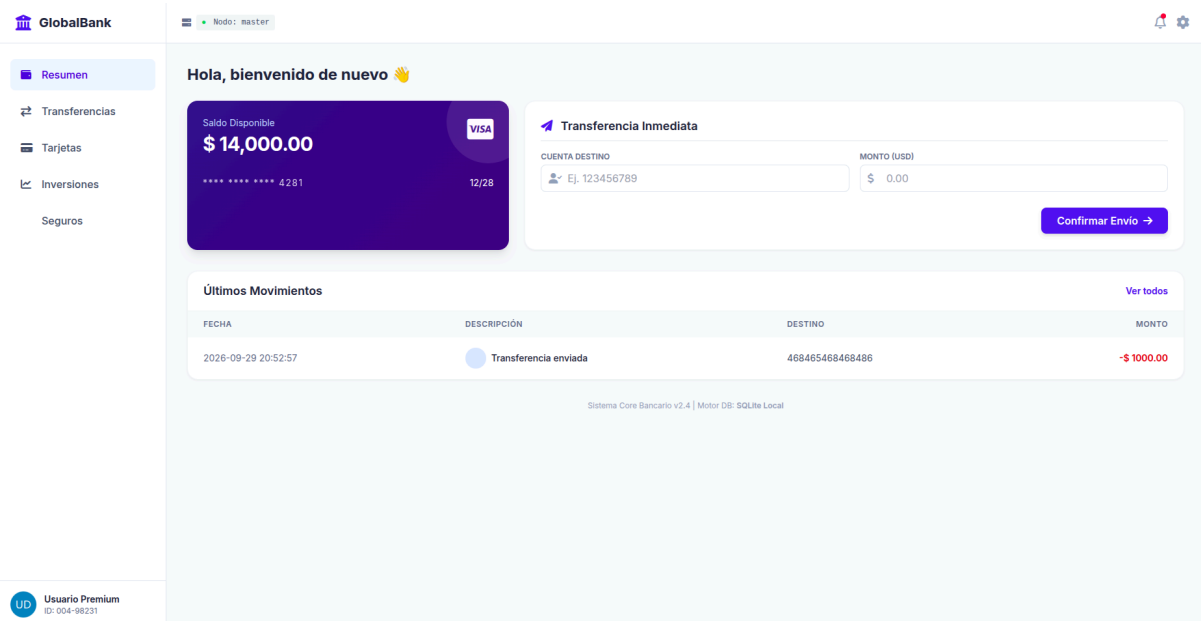
Docente :

Dr. Mamani Aliaga, Alvaro
Henry

Septiembre del 2026



Sistema de banca por internet interfaz



1. Configuración Docker

backend

- Se creó el archivo `Dockerfile` para el backend utilizando Python.
- Creamos el docker file
- Configuración del Dockerfile:

```

prince@master:~/proyecto-banco/backend$ nano Dockerfile
prince@master:~/proyecto-banco/backend$ docker build -t mi-banco-backend:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  26.96MB
Step 1/7 : FROM python:3.10-slim
3.10-slim: Pulling from library/python
6b37362b3da7: Pulling fs layer
7b0c45134812: Pulling fs layer
1f43bcf5ef11: Pulling fs layer
920e618c41a6: Pulling fs layer
920e618c41a6: Waiting
7b0c45134812: Download complete
6b37362b3da7: Verifying Checksum
6b37362b3da7: Download complete
1f43bcf5ef11: Verifying Checksum
1f43bcf5ef11: Download complete
6b37362b3da7: Pull complete
7b0c45134812: Pull complete
1f43bcf5ef11: Pull complete
920e618c41a6: Verifying Checksum
920e618c41a6: Download complete
920e618c41a6: Pull complete
Digest: sha256:9d53d8d4c0e882f61913025db53b3aec4ef74336082a9ab47d8a14e9e8329b00
Status: Downloaded newer image for python:3.10-slim
--> f33084ee4250

```

```

JSON
FROM python:3.10-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app.py .
EXPOSE 3000
CMD ["python", "app.py"]

```

Se procedió a crear la imagen del backend ejecutando el comando: `docker build -t mi-banco-backend:v1 .`

```

prince@master:~/proyecto-banco/backend$ docker build -t mi-banco-backend:v1 .
Collecting itsdangerous>=2.1.2
  Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Collecting MarkupSafe>=2.0
  Downloading markupsafe-3.0.3-cp310-cp310-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (20 kB)
Installing collected packages: psycpg2-binary, MarkupSafe, itsdangerous, click, blinker, Werkzeug, Jinja2, Flask, Flask-CORS
Successfully installed Flask-3.0.0 Flask-CORS-4.0.0 Jinja2-3.1.6 MarkupSafe-3.0.3 Werkzeug-3.1.9 blinker-1.9.0 click-8.1.7 itsdangerous-2.2.0 psycpg2-binary-2.9.9
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv

[notice] A new release of pip is available: 23.0.1 -> 26.2.1
[notice] To update, run: pip install --upgrade pip
--> Removed intermediate container 1aea456b067e
--> 61f6dcf5ae75
Step 5/7 : COPY app.py .
--> 18fe9e596707
Step 6/7 : EXPOSE 3000
--> Running in bee8968af0d1
--> Removed intermediate container bee8968af0d1
--> 2281399ea6fe
Step 7/7 : CMD ["python", "app.py"]
--> Running in 7e6190bb7e78
--> Removed intermediate container 7e6190bb7e78
--> 321a06262992
Successfully built 321a06262992
Successfully tagged mi-banco-backend:v1

```

frontend

- Se creó el archivo Dockerfile para el frontend utilizando Nginx.
- Crearemos el dockerfile
- Configuración del Dockerfile:

```
JSON
FROM nginx:alpine
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
```

Se procedió a crear la imagen del frontend ejecutando el comando: `docker build -t mi-banco-frontend:v1`

```
• prince@master:~/proyecto-banco/frontend$ nano Dockerfile
• prince@master:~/proyecto-banco/frontend$ docker build -t mi-banco-frontend:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon   16.9kB
Step 1/3 : FROM nginx:alpine
alpine: Pulling from library/nginx
e2de96513ba9: Pulling fs layer
d9aae54b5831: Pulling fs layer
6c53d0b2a666: Pulling fs layer
745dfb2690dd: Pulling fs layer
9a9a644fdd6a: Pulling fs layer
64c8194480fe: Pulling fs layer
e76228b47809: Pulling fs layer
e72112c14215: Pulling fs layer
9a9a644fdd6a: Waiting
64c8194480fe: Waiting
e76228b47809: Waiting
745dfb2690dd: Waiting
e72112c14215: Waiting
6c53d0b2a666: Verifying Checksum
6c53d0b2a666: Download complete
d9aae54b5831: Download complete
745dfb2690dd: Download complete
64c8194480fe: Verifying Checksum
64c8194480fe: Download complete
```

```

prince@master:~/proyecto-banco/frontend$ docker build -t mi-banco-frontend:v1 .
e76228b47809: Verifying Checksum
e76228b47809: Download complete
e72112c14215: Verifying Checksum
e72112c14215: Download complete
9a9a644fdd6a: Download complete
e2de96513ba9: Verifying Checksum
e2de96513ba9: Download complete
e2de96513ba9: Pull complete
d9aae54b5831: Pull complete
6c53d0b2a666: Pull complete
745dfb2690dd: Pull complete
9a9a644fdd6a: Pull complete
64c8194480fe: Pull complete
e76228b47809: Pull complete
e72112c14215: Pull complete
Digest: sha256:df221db836e1754089190208cee7eeda94f233197056426eda74a43ab1abeac2
Status: Downloaded newer image for nginx:alpine
--> 3dd08163706a
Step 2/3 : COPY index.html /usr/share/nginx/html/index.html
--> 895a0544382c
Step 3/3 : EXPOSE 80
--> Running in c4ed2148993c
--> Removed intermediate container c4ed2148993c
--> 676378260d54
Successfully built 676378260d54
Successfully tagged mi-banco-frontend:v1
prince@master:~/proyecto-banco/frontend$

```

2. Configuración Kubernetes

Se crearon los manifiestos YAML en la carpeta **k8s** para el despliegue de la arquitectura:

1-postgres.yaml

```

JSON
apiVersion: apps/v1
kind: Deployment
metadata:
  name: postgres-db
spec:
  replicas: 1
  selector:
    matchLabels:
      app: postgres-db
  template:
    metadata:
      labels:
        app: postgres-db
    spec:
      containers:
        - name: postgres
          image: postgres:15-alpine
          env:
            - name: POSTGRES_USER
              value: "postgres"

```

```

      - name: POSTGRES_PASSWORD
        value: "password"
      - name: POSTGRES_DB
        value: "banco_db"
    ports:
      - containerPort: 5432
---
apiVersion: v1
kind: Service
metadata:
  name: postgres-service
spec:
  selector:
    app: postgres-db
  ports:
    - port: 5432

```

2-backend.yaml

```

JSON
apiVersion: apps/v1
kind: Deployment
metadata:
  name: backend-banco
spec:
  replicas: 2
  selector:
    matchLabels:
      app: backend-banco
  template:
    metadata:
      labels:
        app: backend-banco
    spec:
      containers:
        - name: backend
          image: mi-banco-backend:v1
          imagePullPolicy: Never
          env:
            - name: DB_HOST
              value: "postgres-service"
          ports:

```

```

      - containerPort: 3000
    resources:
      requests:
        cpu: "50m"
      limits:
        cpu: "250m"
---
apiVersion: v1
kind: Service
metadata:
  name: backend-service
spec:
  selector:
    app: backend-banco
  ports:
    - port: 3000
---
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: backend-hpa
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: backend-banco
  minReplicas: 2
  maxReplicas: 10
  metrics:
    - type: Resource
      resource:
        name: cpu
        target:
          type: Utilization
          averageUtilization: 50

```

3-frontend.yaml

```

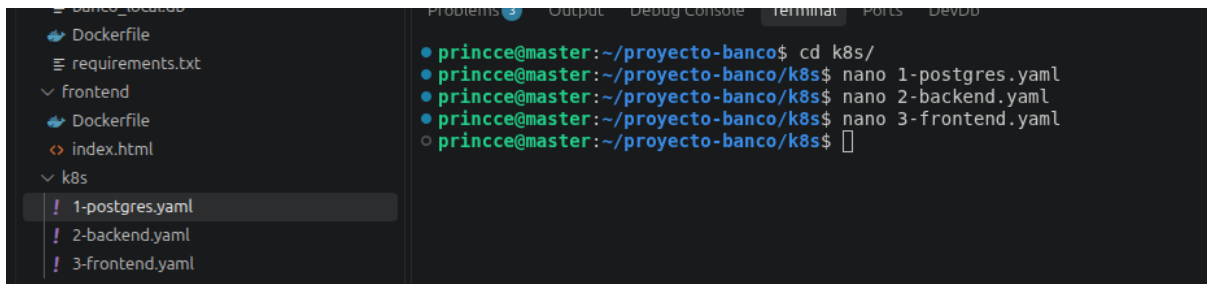
JSON
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend-banco
spec:
  replicas: 1
  selector:

```

```

    matchLabels:
      app: frontend-banco
  template:
    metadata:
      labels:
        app: frontend-banco
    spec:
      containers:
        - name: frontend
          image: mi-banco-frontend:v1
          imagePullPolicy: Never
          ports:
            - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: frontend-service
spec:
  type: NodePort
  selector:
    app: frontend-banco
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30080

```



3. Clúster Kind y Despliegue

- **Creación del Clúster:**

- Se procedió a crear el nuevo clúster en Kind asignándole el nombre "banco".
- Comando ejecutado: `kind create cluster --name banco`.


```

prince@master:~/proyecto-banco$ kind create cluster --name banco
Creating cluster "banco" ...
  ✓ Ensuring node image (kindest/node:v1.37.0)
  ✓ Preparing nodes
  ✓ Writing configuration
  ✓ Starting control-plane
  ✓ Installing CNI
  ✓ Installing StorageClass
Set kubectl context to "kind-banco"
You can now use your cluster with:

kubectl cluster-info --context kind-banco

Not sure what to do next? Check out https://kind.sigs.k8s.io/docs/user/quick-start/

```

- **Instalación del Sensor de Métricas (Metrics Server):**

- Se instaló manualmente y se le aplicó un parche de seguridad al correr en un entorno local.
- Comando de instalación: `kubectl apply -f [https://github.com/kubernetes-sigs/metrics-server/releases/latest/download/components.yaml](https://github.com/kubernetes-sigs/metrics-server/releases/latest/download/components.yaml)`.
- Comando para parchear la seguridad local: `kubectl patch deployment metrics-server -n kube-system --type 'json' -p '[{"op": "add", "path": "/spec/template/spec/containers/0/args/-", "value": "--kubelet-insecure-tls"}]'`.

```

prince@master:~/proyecto-banco$ kubectl apply -f https://github.com/kubernetes-sigs/metrics-server/releases/latest/download/components.yaml
serviceaccount/metrics-server created
clusterrole.rbac.authorization.k8s.io/system:aggregated-metrics-reader created
clusterrole.rbac.authorization.k8s.io/system:metrics-server created
rolebinding.rbac.authorization.k8s.io/metrics-server-auth-reader created
clusterrolebinding.rbac.authorization.k8s.io/metrics-server:system:auth-delegator created
clusterrolebinding.rbac.authorization.k8s.io/system:metrics-server created
service/metrics-server created
deployment.apps/metrics-server created
apiservice.apiregistration.k8s.io/v1beta1.metrics.k8s.io created

prince@master:~/proyecto-banco$ kubectl patch deployment metrics-server -n kube-system --type 'json' -p '[{"op": "add", "path": "/spec/template/spec/containers/0/args/-", "value": "--kubelet-insecure-tls"}]'
deployment.apps/metrics-server patched

```

- **Carga de las imágenes Docker al clúster:**

- `kind load docker-image mi-banco-backend:v1 --name banco`
- `kind load docker-image mi-banco-frontend:v1 --name banco`

```

prince@master:~/proyecto-banco$ kind load docker-image mi-banco-backend:v1 --name banco
Image: "mi-banco-backend:v1" with ID "sha256:321a062629927b5f0a854db68e69cf107632e8202363ac873efe9370c2d47dc9" not yet present on node "banco-control-plane", loading...
prince@master:~/proyecto-banco$ kind load docker-image mi-banco-frontend:v1 --name banco
Image: "mi-banco-frontend:v1" with ID "sha256:676378260d54282714011e60d219b8d4e749acc4d5291b4ef0e5ee22a0ea25f3" not yet present on node "banco-control-plane", loading...
prince@master:~/proyecto-banco$

```

- **Despliegue de la arquitectura:**

- Se aplicaron todos los archivos YAML ejecutando: `kubectl apply -f k8s/`.

```

● prince@master:~/proyecto-banco$ kubectl apply -f k8s/
deployment.apps/postgres-db created
service/postgres-service created
deployment.apps/backend-banco created
service/backend-service created
horizontalpodautoscaler.autoscaling/backend-hpa created
deployment.apps/frontend-banco created
service/frontend-service created
○ prince@master:~/proyecto-banco$ █

```

- **Apertura de la web mediante Port-Forwarding:**

- Para la interfaz web: kubectl port-forward svc/frontend-service 8080:80.
- Para el backend: kubectl port-forward svc/backend-service 3000:3000.

```

○ prince@master:~/proyecto-banco$ kubectl port-forward svc/frontend-service 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
█

```

```

○ prince@master:~/proyecto-banco$ kubectl port-forward svc/backend-service 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
█

```

4. Instalación de Herramientas de Monitoreo y Pruebas

- **Instalación de K9s:** curl -sS https://webinstall.dev/k9s | bash.
- **Instalación de K6:**
 - Descarga: curl -L https://github.com/grafana/k6/releases/download/v0.47.0/k6-v0.47.0-linux-amd64.tar.gz -o k6.tar.gz.
 - Extracción: tar -xzf k6.tar.gz.
 - Movimiento a binarios: sudo mv k6-v0.47.0-linux-amd64/k6 /usr/local/bin/k6.

```

prince@master:~/proyecto-banco$ curl -sS https://webinstall.dev/k9s | bash

>>> Welcome to Webi! - modern tools, instant installs. <<<
We expect your experience to be absolutely perfect!

Success? Star it! https://github.com/webinstall/webi-installers
Problem? Report it: https://github.com/webinstall/webi-installers/issues
                    (your system is GNU/Linux/x86_64 with libc & curl+wget)

Bootstrapping Webi
  Downloading https://webinstall.dev/packages/webi/webi.sh
    to ~/.local/bin/webi
  Running ~/.local/bin/webi k9s@stable

Installing k9s ...
  Found ~/.local/bin
  Initializing ~/.config/envman/
  Edit ~/.profile to source ~/.config/envman/load.sh
  Edit ~/.bashrc to source ~/.config/envman/load.sh

```

```

prince@master:~/proyecto-banco$ curl -L https://github.com/grafana/k6/releases/download/v0.47.0/k6-v0.47.0-linux-amd64.tar.gz -o k6.tar.gz
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
  0     0    0     0    0     0      0      0  0:00:01  0:00:01  0:00:01  25.4M
100 22.8M 100 22.8M    0     0  14.0M      0  0:00:01  0:00:01  0:00:01 25.4M
prince@master:~/proyecto-banco$ tar -xzf k6.tar.gz
prince@master:~/proyecto-banco$ sudo mv k6-v0.47.0-linux-amd64/k6 /usr/local/bin/k6
[sudo] contraseña para prince:

```

5. Primera Prueba de Estrés e Incidentes

- **Análisis del Cuello de Botella en la Red Local (Port-Forwarding):**
 - Durante la primera fase de las pruebas de estrés con K6, experimentamos una pérdida temporal de disponibilidad en la interfaz web de la aplicación (la página dejó de mostrar los datos financieros y se quedó en estado de carga).
 - Además, la herramienta K6 comenzó a registrar múltiples errores de tipo request timeout.
 - Al realizar el diagnóstico del incidente, el equipo determinó que el clúster de Kubernetes y el Autoescalador (HPA) funcionaban correctamente y no habían colapsado. El fallo se originó en la capa de red utilizada para la prueba.

42 metadata:

Problem Output Debug Console Terminal Ports DevDb

Context: kind-banco [RW]
Cluster: kind-banco
User: kind-banco
K9s Rev: v0.51.0
K8s Rev: v1.37.0
CPU: 2%
MEM: 6%

<0> all
<1> default

<a> Attach
<ctrl-d> Delete
<d> Describe
<e> Edit
<f> Help
<shift-j> Jump Owner

<ctrl-k> Kill
<l> Logs
<p> Logs Previous
<shift-f> Port-Forward
<z> Sanitize
<s> Shell

<o> Show Node
<f> Show PortForward
<t> Transfer
<w> Warp To Namespace
<y> YAML



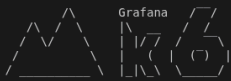
NAME	PF	READY	STATUS	RESTARTS	pods(default) [13]			MEM	%MEM/R	%MEM/L	IP	NODE	AGE
					CPU	%CPU/R	%CPU/L						
backend-banco-57b55b56c-6mtlp	●	1/1	Running	0	1	2	0	22	n/a	n/a	10.244.0.12	banco-control-plane	11m
backend-banco-57b55b56c-6w6z4	●	1/1	Running	2	250	500	100	32	n/a	n/a	10.244.0.8	banco-control-plane	32m
backend-banco-57b55b56c-b9rsb	●	1/1	Running	0	1	2	0	21	n/a	n/a	10.244.0.15	banco-control-plane	11m
backend-banco-57b55b56c-fqkfz	●	1/1	Running	0	1	2	0	21	n/a	n/a	10.244.0.11	banco-control-plane	11m
backend-banco-57b55b56c-hrhd4	●	1/1	Running	0	1	2	0	21	n/a	n/a	10.244.0.20	banco-control-plane	10m
backend-banco-57b55b56c-jks7w	●	1/1	Running	0	1	2	0	21	n/a	n/a	10.244.0.19	banco-control-plane	10m
backend-banco-57b55b56c-nbscb	●	1/1	Running	0	1	2	0	21	n/a	n/a	10.244.0.17	banco-control-plane	10m
backend-banco-57b55b56c-qjm7k	●	1/1	Running	0	1	2	0	21	n/a	n/a	10.244.0.16	banco-control-plane	11m
backend-banco-57b55b56c-r5pzd	●	1/1	Running	0	1	2	0	21	n/a	n/a	10.244.0.18	banco-control-plane	10m
backend-banco-57b55b56c-rsvcg	●	1/1	Running	0	1	2	0	21	n/a	n/a	10.244.0.14	banco-control-plane	11m
backend-banco-57b55b56c-tbnpj	●	1/1	Running	2	1	2	0	21	n/a	n/a	10.244.0.9	banco-control-plane	32m
frontend-banco-6f96f6cc58-8qbdb	●	1/1	Running	0	0	n/a	n/a	13	n/a	n/a	10.244.0.10	banco-control-plane	32m
postgres-db-696c5685d9-6j9x6	●	1/1	Running	0	1	n/a	n/a	25	n/a	n/a	10.244.0.7	banco-control-plane	32m

- Para medir la verdadera capacidad de procesamiento y autoescalado de nuestros contenedores sin interferencias de la red local, decidimos cambiar la estrategia del ataque. Modificamos el vector de inyección para ejecutar el contenedor de K6 directamente desde adentro del clúster de Kubernetes.
- Al hacer esto, el ataque se comunicó de forma nativa a través de la red interna de Kubernetes (apuntando al backend-service), eliminando el cuello de botella del túnel y permitiendo que la aplicación web (Frontend) siguiera consumiendo datos con total normalidad a través del puerto liberado, incluso mientras el sistema recibía una carga masiva.
- **Comando ejecutado para el ataque interno:** `kubectl run k6-ataque --image=grafana/k6 -i --rm --restart=Never -- run - < ataque.js`.

```
Context: kind-banco [RW]
Cluster: kind-banco
User: kind-banco
K9s Rev: v0.51.0
K8s Rev: v1.37.0
CPU: 1%
MEM: 7%
```

NAME	PF	READY	STATUS	RESTARTS	CPU	%CPU/R	%CPU/L	MEM	%MEM/R	%MEM/L	IP	NODE	AGE
backend-banco-54ddc48ffc-9vxlz	●	1/1	Running	0	1	2	0	21	n/a	n/a	10.244.0.27	banco-control-plane	3m5s
backend-banco-54ddc48ffc-lq7zv	●	1/1	Running	0	1	2	0	21	n/a	n/a	10.244.0.28	banco-control-plane	3m4s
frontend-banco-6f96f6cc58-8qbdb	●	1/1	Running	0	0	n/a	n/a	13	n/a	n/a	10.244.0.10	banco-control-plane	42m
postgres-db-696c5685d9-6j9x6	●	1/1	Running	0	1	n/a	n/a	25	n/a	n/a	10.244.0.7	banco-control-plane	42m

```
prince@master:~/proyecto-banco$ kubectl run k6-ataque --image=grafana/k6 -i --rm --restart=Never -- run - < ataque.js
All commands and output from this session will be recorded in container logs, including credentials and sensitive information passed through the command prompt.
If you don't see a command prompt, try pressing enter.
```



```

execution: local
script: -
output: -

scenarios: (100.00%) 1 scenario, 150 max VUs, 2m20s max duration (incl. graceful stop):
  * default: Up to 150 looping VUs for 1m50s over 3 stages (gracefulRampDown: 30s, gracefulStop: 30s)

running (0m01.0s), 002/150 VUs, 0 complete and 0 interrupted iterations
default [ 1% ] 002/150 VUs 0m01.0s/1m50.0s

running (0m02.0s), 004/150 VUs, 0 complete and 0 interrupted iterations
default [ 2% ] 004/150 VUs 0m02.0s/1m50.0s

running (0m03.0s), 005/150 VUs, 0 complete and 0 interrupted iterations
default [ 3% ] 005/150 VUs 0m03.0s/1m50.0s
```

GlobalBank

Resumen

Transferencias

Tarjetas

Inversiones

Seguros

Procesado por: backend-banco-54ddc48ffc-9vx1z

Hola, bienvenido de nuevo 🙌

Saldo Disponible

\$13,900.00

VISA

**** * 4281

12/28

Transferencia Inmediata

CUENTA DESTINO

Ej. 123456789

MONTO (USD)

\$ 0.00

Transferencia Exitosa!

Confirmar Envío →

Últimos Movimientos

Ver todos

FECHA	DESCRIPCIÓN	DESTINO	MONTO
2026-09-30 04:51	Transferencia enviada	3086468	\$1000.00
2026-09-30 04:50	Transferencia enviada	1000	\$100.00

Sistema Core Bancario v2.4 | Motor DB: PostgreSQL

1 import http from 'k6/http';
2 import { check, sleep } from 'k6';
3
4 export let options = {
5 stages: [
6 { duration: '30s', target: 50 },
7 { duration: '1m', target: 150 },
8 { duration: '20s', target: 0 },
9],
10 };
11
12 export default function () {
13 let res = http.get('http://backend-service:3000/api/stress');
14 check(res, { 'Servidor responde': () => r.status == 200 });
15 sleep(0.1);
16 }

Context: kind-banco [RW]
Cluster: kind-banco
User: kind-banco
K8s Rev: v0.51.0
K8s Rev: v1.37.0
CPU: 17%
MEM: 8%

pod

NAME	IP	READY	STATUS	RESTARTS	CPU	MEM
backend-banco-54ddc48ffc-2jhv		1/1	Running	0	183	286
backend-banco-54ddc48ffc-4b2ix		1/1	Running	0	47	0
backend-banco-54ddc48ffc-4m9v		1/1	Running	0	207	413
backend-banco-54ddc48ffc-4qhzh		1/1	Running	0	213	426
backend-banco-54ddc48ffc-5jdrs		1/1	Running	0	0	0
backend-banco-54ddc48ffc-6h4m		1/1	Running	0	33	66
backend-banco-54ddc48ffc-6wdgm		1/1	Running	0	0	0
backend-banco-54ddc48ffc-7w9m		1/1	Running	0	0	0
backend-banco-54ddc48ffc-9vx1z		1/1	Running	0	251	501
backend-banco-54ddc48ffc-57z6t		1/1	Running	0	0	0
backend-banco-54ddc48ffc-6dnk2		1/1	Running	0	0	0
backend-banco-54ddc48ffc-94dcn		1/1	Running	0	57	113
backend-banco-54ddc48ffc-96rnt		1/1	Running	0	0	0
backend-banco-54ddc48ffc-b5qst		1/1	Running	0	0	0
backend-banco-54ddc48ffc-c2b4		1/1	Running	0	0	0
backend-banco-54ddc48ffc-dmth9		1/1	Running	0	251	501
backend-banco-54ddc48ffc-fvzr2		1/1	Running	0	203	405

GlobalBank

Resumen

Transferencias

Tarjetas

Inversiones

Seguros

Nodo: backend-banco-54ddc48ffc-9vx1z

Hola, bienvenido de nuevo 🙌

Saldo Disponible

\$8,900.00

VISA

**** * 4281

12/28

Transferencia Inmediata

CUENTA DESTINO

Ej. 123456789

MONTO (USD)

\$ 0.00

Confirmar Envío →

Últimos Movimientos

Ver todos

FECHA	DESCRIPCIÓN	DESTINO	MONTO
2026-09-30 04:51	Transferencia enviada	4564645	\$5000.00
2026-09-30 04:51	Transferencia enviada	3086468	\$1000.00
2026-09-30 04:50	Transferencia enviada	1000	\$100.00

1 import http from 'k6/http';
2 import { check, sleep } from 'k6';
3
4 export let options = {
5 stages: [
6 { duration: '30s', target: 50 },
7 { duration: '1m', target: 150 },
8 { duration: '20s', target: 0 },
9],
10 };
11
12 export default function () {
13 let res = http.get('http://backend-service:3000/api/stress');
14 check(res, { 'Servidor responde': () => r.status == 200 });
15 sleep(0.1);
16 }

Context: kind-banco [RW]
Cluster: kind-banco
User: kind-banco
K8s Rev: v0.51.0
K8s Rev: v1.37.0
CPU: 46%
MEM: 11%

pod

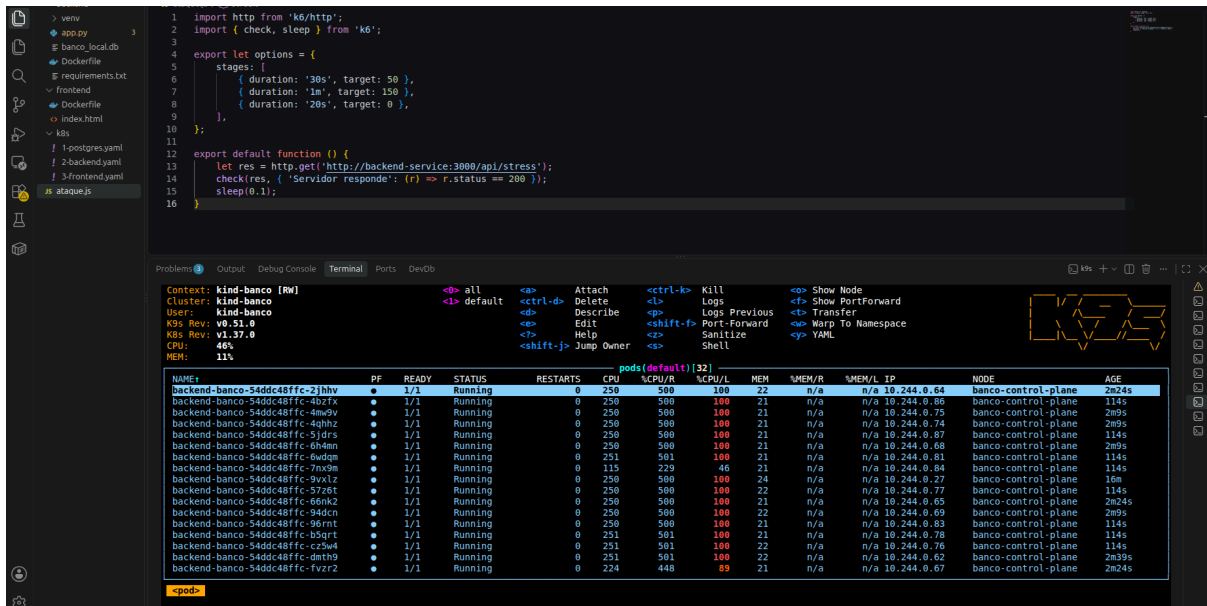
NAME	IP	READY	STATUS	RESTARTS	CPU	MEM
backend-banco-54ddc48ffc-fvzr2		1/1	Running	0	224	448
backend-banco-54ddc48ffc-gcdmw		1/1	Running	0	250	500
backend-banco-54ddc48ffc-gxzf6		1/1	Running	0	250	500
backend-banco-54ddc48ffc-j429h		1/1	Running	0	251	501
backend-banco-54ddc48ffc-kbwgx		1/1	Running	0	251	501
backend-banco-54ddc48ffc-l97sv		1/1	Running	0	250	500
backend-banco-54ddc48ffc-nllqr		1/1	Running	0	251	501
backend-banco-54ddc48ffc-pl4k9		1/1	Running	0	251	501
backend-banco-54ddc48ffc-pll7m		1/1	Running	0	215	430
backend-banco-54ddc48ffc-pn5m4		1/1	Running	0	251	501
backend-banco-54ddc48ffc-q6sbw		1/1	Running	0	250	500
backend-banco-54ddc48ffc-r5dsw		1/1	Running	0	250	500
backend-banco-54ddc48ffc-sgd28		1/1	Running	0	251	501
backend-banco-54ddc48ffc-vb6z2		1/1	Running	0	250	500
frontend-banco-6f96fec38-8qdbd		1/1	Running	0	1	n/a
k6-ataque		1/1	Running	0	1	n/a
postgres-db-696c585d9-6j9x6		1/1	Running	0	1	n/a

1 import http from 'k6/http';
2 import { check, sleep } from 'k6';
3
4 export let options = {
5 stages: [
6 { duration: '30s', target: 50 },
7 { duration: '1m', target: 150 },
8 { duration: '20s', target: 0 },
9],
10 };
11
12 export default function () {
13 let res = http.get('http://backend-service:3000/api/stress');
14 check(res, { 'Servidor responde': () => r.status == 200 });
15 sleep(0.1);
16 }

Context: kind-banco [RW]
Cluster: kind-banco
User: kind-banco
K8s Rev: v0.51.0
K8s Rev: v1.37.0
CPU: 46%
MEM: 11%

pod

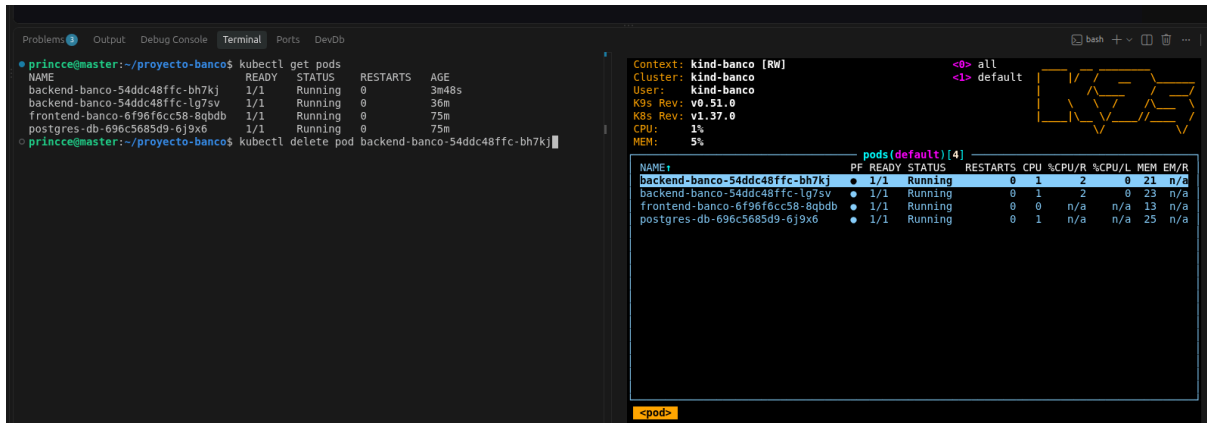
NAME	IP	READY	STATUS	RESTARTS	CPU	%CPU/R	%CPU/L	MEM	%MEM/R	%MEM/L	IP	NODE	AGE
backend-banco-54ddc48ffc-dmth9		1/1	Running	0	251	501	100	22	n/a	n/a	10.244.0.62	banco-control-plane	2m29s
backend-banco-54ddc48ffc-fvzr2		1/1	Running	0	224	448	89	21	n/a	n/a	10.244.0.67	banco-control-plane	2m14s
backend-banco-54ddc48ffc-gcdmw		1/1	Running	0	250	500	100	22	n/a	n/a	10.244.0.63	banco-control-plane	2m29s
backend-banco-54ddc48ffc-gxzf6		1/1	Running	0	250	500	100	21	n/a	n/a	10.244.0.82	banco-control-plane	1m4s
backend-banco-54ddc48ffc-j429h		1/1	Running	0	251	501	100	22	n/a	n/a	10.244.0.70	banco-control-plane	119s
backend-banco-54ddc48ffc-kbwgx		1/1	Running	0	251	501	100	21	n/a	n/a	10.244.0.88	banco-control-plane	1m4s
backend-banco-54ddc48ffc-l97sv		1/1	Running	0	250	500	100	24	n/a	n/a	10.244.0.28	banco-control-plane	16m
backend-banco-54ddc48ffc-nllqr		1/1	Running	0	251	501	100	22	n/a	n/a	10.244.0.66	banco-control-plane	2m14s
backend-banco-54ddc48ffc-pl4k9		1/1	Running	0	251	501	100	21	n/a	n/a	10.244.0.71	banco-control-plane	119s
backend-banco-54ddc48ffc-pll7m		1/1	Running	0	215	430	86	21	n/a	n/a	10.244.0.79	banco-control-plane	1m4s
backend-banco-54ddc48ffc-pn5m4		1/1	Running	0	251	501	100	22	n/a	n/a	10.244.0.89	banco-control-plane	1m4s
backend-banco-54ddc48ffc-q6sbw		1/1	Running	0	250	500	100	22	n/a	n/a	10.244.0.72	banco-control-plane	119s
backend-banco-54ddc48ffc-r5dsw		1/1	Running	0	250	500	100	21	n/a	n/a	10.244.0.85	banco-control-plane	1m4s
backend-banco-54ddc48ffc-sgd28		1/1	Running	0	251	501	100	22	n/a	n/a	10.244.0.80	banco-control-plane	1m4s
backend-banco-54ddc48ffc-vb6z2		1/1	Running	0	250	500	100	21	n/a	n/a	10.244.0.73	banco-control-plane	119s
frontend-banco-6f96fec38-8qdbd		1/1	Running	0	1	n/a	n/a	13	n/a	n/a	10.244.0.10	banco-control-plane	56m
postgres-db-696c585d9-6j9x6		1/1	Running	0	1	n/a	n/a	25	n/a	n/a	10.244.0.7	banco-control-plane	56m

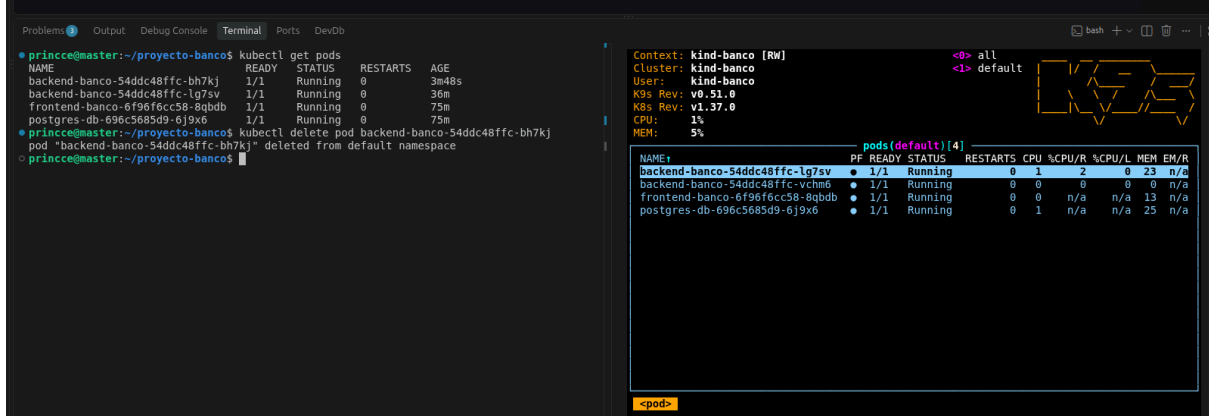
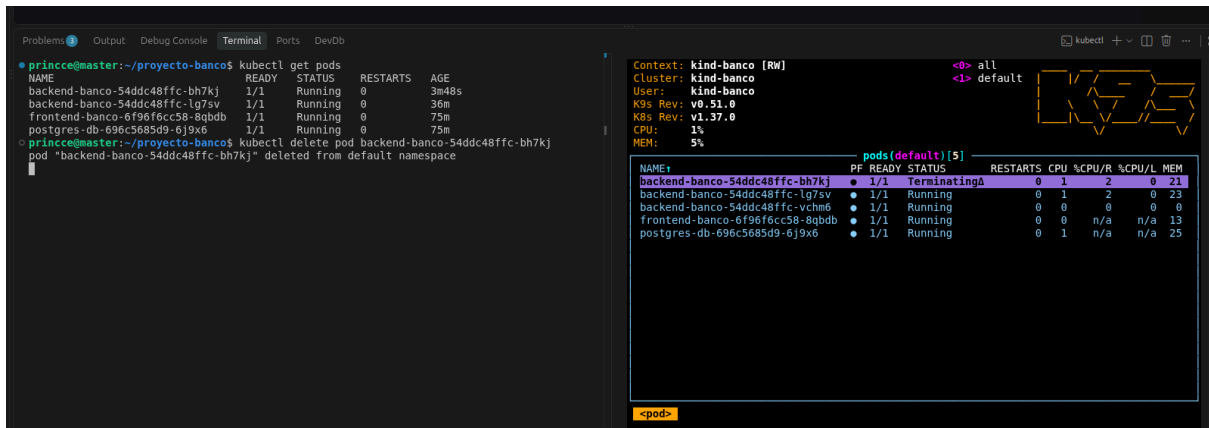


7. Prueba de Tolerancia a Fallas

- Eliminación de un pod desde la terminal:
 - Se listaron los contenedores con el comando `kubectl get pods`.
 - Se procedió a eliminar uno utilizando: `kubectl delete pod backend-banco-54ddc48ffc-bh7kj`.

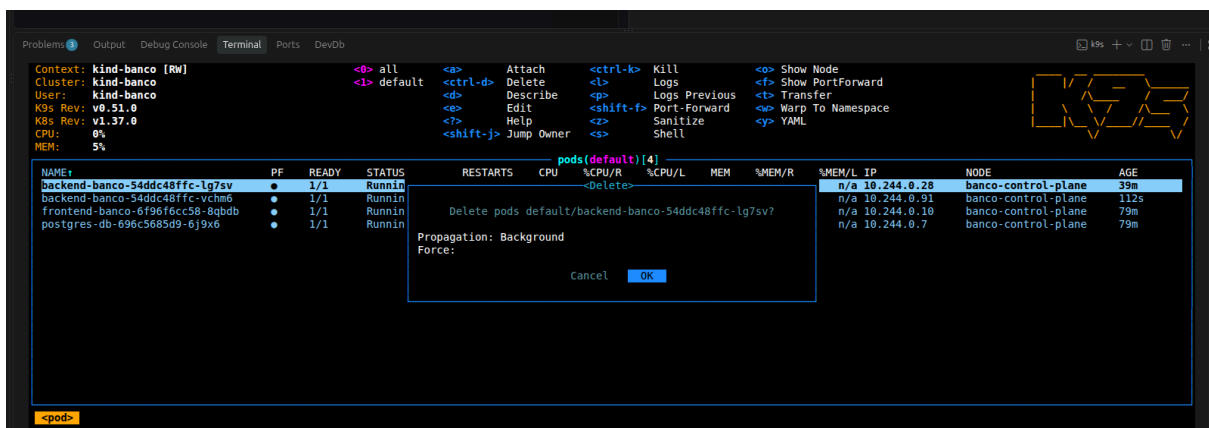
Eliminado un pod desde la terminal





● Eliminación de un pod desde K9s:

- También podemos eliminar pods directamente desde la herramienta visual K9s.
- Para ello, seleccionamos el contenedor y presionamos control-d (Delete).
- Se confirmó la acción en el cuadro de diálogo y el sistema procedió de inmediato a cambiar el estado del pod a Terminating, creando automáticamente un contenedor de reemplazo



ProblemsOutputDebug ConsoleTerminalPortsDevDb

Context: kind-banco [RW]
Cluster: kind-banco
User: kind-banco
K9s Rev: v0.51.0
K8s Rev: v1.37.0
CPU: 0%
MEM: 5%

<0> all
<1> default
<ctrl-d> Attach
<ctrl-d> Delete
<ctrl-d> Describe
<ctrl-d> Edit
<ctrl-d> Help
<shift-j> Jump Owner

<ctrl-k> Kill
<ctrl-k> Logs
<ctrl-k> Logs Previous
<ctrl-k> Port-Forward
<ctrl-k> Sanitize
<ctrl-k> Shell

<0> Show Node
<f> Show PortForward
<t> Transfer
<w> Warp To Namespace
<y> YAML

NAME

PF

READY

STATUS

RESTARTS

CPU

%CPU/R

%CPU/L

MEM

%MEM/R

%MEM/L

IP

NODE

AGE

backend-banco-54ddc48ffc-jf5nv

●

0/1

ContainerCreating

0

0

0

0

0

n/a

n/a

n/a

backend-banco-54ddc48ffc-lg7sv

●

1/1

Terminating

0

1

2

0

23

n/a

n/a

10.244.0.28

backend-banco-54ddc48ffc-vchm6

●

1/1

Running

0

1

2

0

21

n/a

n/a

10.244.0.91

frontend-banco-6f96f6cc58-8qbdb

●

1/1

Running

0

0

n/a

n/a

13

n/a

n/a

10.244.0.10

postgres-db-696c5685d9-6j9x6

●

1/1

Running

0

1

n/a

n/a

25

n/a

n/a

10.244.0.7

banco-control-plane

4s

banco-control-plane

49m

banco-control-plane

2m13s

banco-control-plane

79m

banco-control-plane

79m

<pod>

Delete resource v1/pods default/backend-banco-54ddc48ffc-lg7sv

ProblemsOutputDebug ConsoleTerminalPortsDevDb

Context: kind-banco [RW]
Cluster: kind-banco
User: kind-banco
K9s Rev: v0.51.0
K8s Rev: v1.37.0
CPU: 1%
MEM: 5%

<0> all
<1> default
<ctrl-d> Attach
<ctrl-d> Delete
<ctrl-d> Describe
<ctrl-d> Edit
<ctrl-d> Help
<shift-j> Jump Owner

<ctrl-k> Kill
<ctrl-k> Logs
<ctrl-k> Logs Previous
<ctrl-k> Port-Forward
<ctrl-k> Sanitize
<ctrl-k> Shell

<0> Show Node
<f> Show PortForward
<t> Transfer
<w> Warp To Namespace
<y> YAML

NAME

PF

READY

STATUS

RESTARTS

CPU

%CPU/R

%CPU/L

MEM

%MEM/R

%MEM/L

IP

NODE

AGE

backend-banco-54ddc48ffc-jf5nv

●

1/1

Running

0

0

0

0

0

n/a

n/a

10.244.0.92

backend-banco-54ddc48ffc-lg7sv

●

1/1

Terminating

0

1

2

0

23

n/a

n/a

10.244.0.28

backend-banco-54ddc48ffc-vchm6

●

1/1

Running

0

1

2

0

21

n/a

n/a

10.244.0.91

frontend-banco-6f96f6cc58-8qbdb

●

1/1

Running

0

0

n/a

n/a

13

n/a

n/a

10.244.0.10

postgres-db-696c5685d9-6j9x6

●

1/1

Running

0

1

n/a

n/a

25

n/a

n/a

10.244.0.7

banco-control-plane

4s

banco-control-plane

49m

banco-control-plane

2m17s

banco-control-plane

79m

banco-control-plane

79m

<pod>